Ethan K. Sun

ethan.k.sun.26@dartmouth.edu | GitHub

Education

Dartmouth College **Hanover, NH**
Bachelor of Arts - Economics, High Honors *September 2022 - Present*

- GPA: 3.78 / 4.00
- SAT: 1580 (Math: 800, Reading/Writing: 780); SAT Subject Test Math Level II: 800
- Brother of the Psi Epsilon Chapter of Zeta Psi

Experience

Research Assistant **Chicago, IL**
Federal Reserve Bank of Chicago *June 2026 - Present*

- Provide support to the economists on the Financial Economics Team in the Research, Policy & Public Engagement Department at the Federal Reserve Bank of Chicago
- Collect and clean data, perform statistical and economic analysis, create figures, write code, edit papers, and report results

Stata Teaching Assistant **Hanover, NH**
Social Science Data Lab at Dartmouth College *Winter 2026*

- Held weekly office hours for students taking their culminating experience course in economics and assist with Stata syntax and statistical inference

Economics Teaching Assistant **Hanover, NH**
Faculty - Maura Doyle, Diego Ramos-Toro *Fall 2024, Spring 2025, Fall 2025*

- ECON 01 (The Price System: Analysis, Problems, and Policies) - Graded problem sets, exams, and held weekly office hours for a class of 90 students to ask questions about the course material
- ECON 21 (Microeconomics) - Held three hours of weekly office hours for a class of 60 students to ask questions about the course material

President of Dartmouth Economic Research Scholars **Hanover, NH**
Department of Economics at Dartmouth College *April 2025 - March 2026*

- Serve as a liaison between DERS students, faculty, and alumni. Organize and facilitate meetings, coordinate events, and secure funding for panels, colloquiums, workshops, and distinguished guest lectures to promote undergraduate economic research and connect students with professors and industry professionals

Economics Research Assistant **Hanover, NH**
Principal Investigator - Erzo F.P. Luttmer *September 2024 - March 2025*

- Awarded Dartmouth Economic Research Scholar and James O. Freedman Presidential Scholar with $3,200 grant for this role. Worked on The Dartmouth Professor in Economics Erzo F.P. Luttmer's team to value human lives based on respondents' choice between vignettes of life stories where life-time income and age of death are randomized
- Created literature reviews, tested survey data to ensure randomizations were implemented correctly and that respondent responses were coded accurately, produced high-quality exhibits based on data analyzed by Professor Luttmer or other senior members of the research team, and helped proofread, edit, and format the paper

Pharmaceutical Engineering Research Assistant **Hanover, NH**
Principal Investigator - Xin Qi *March 2023 - November 2023*

- Awarded URAD Scholar and William H. Neukom 1964 Scholar with $2,400 grant for this role. Worked in Assistant Professor Xin Qi's Qi Lab at Dartmouth and collaborated with pharmaceutical companies to develop antibodies to target Interleukin-33, a protein in the IL-1 family of cytokines known to be associated with various inflammatory and allergic conditions

- Analyzed data by utilizing RCSB, VMD, Pymol, and AlphaFold to visualize and analyze the structure of antibodies targeting Interleukin-33, then used GROMACS, Slurm, and Python to perform molecular dynamic simulations for data analysis and studying of their properties and binding efficacy to Interleukin-33

Publications & Presentations

- Sun, Ethan. "There Goes the Neighborhood: How Gun Violence Affects Housing Prices." Available at SSRN 6065666 (November 26th, 2025). [pdf]
- Sun, Ethan. "Long-Term Effects of the ACA Medicaid Expansions." Available at SSRN 5197610 (March 15th, 2025). [pdf]
- Sun, Alan, **Ethan Sun**, and Warren Shepard. "Algorithmic Phase Transitions in Large Language Models: A Mechanistic Case Study of Arithmetic." Paper presented at ATTRIB @ NeurIPS 2024 workshop, Vancouver, BC, Canada (December 14th, 2024). [pdf]

Honors & Awards

Third Honor Group **Hanover, NH**
Dartmouth College *June 2026*

- Awarded to students graduating within the top 35% of undergraduate GPAs

Economics Department Service and Leadership Award **Hanover, NH**
Department of Economics at Dartmouth College *June 2026*

- Awarded to students who have made significant contributions to the Economics Department through TA, RA, or other work

James O. Freedman Presidential Scholar **Hanover, NH**
Dartmouth College Undergraduate Advising and Research *May 2024*

- Prestigious, GPA-based two-term undergraduate research program with $3,200 grant

Dartmouth Economic Research Scholar **Hanover, NH**
Dartmouth College Department of Economics *March 2024*

- Competitive program designed to support Presidential Scholars who are majoring in economics and provides additional mentoring and support to students with strong potential in economics research

William H. Neukom 1964 Scholar **Hanover, NH**
The Neukom Institute for Computational Science *August 2023*

- Selective program that provides a $1,600 ($1,200 at time of award) grant to enrolled undergraduates engaged in faculty-advised research in the development of novel computational techniques as well as the application of computational methods to problems in the Sciences, Social Sciences, Humanities, and the Arts

Undergraduate Research Assistantships at Dartmouth Scholar **Hanover, NH**
Dartmouth College Undergraduate Advising and Research *March 2023*

- Competitive program that provides a $1,600 ($1,200 at time of award) grant intended to introduce students to research with priority given to sophomores and juniors (received award as a freshman)

Citations for Meritorious Performance **Hanover, NH**
Dartmouth College *Spring 2023, Summer 2024*

- Probability, Research Associate Linh Huynh
- Introduction to Programming and Computation, Albert Bradley 1915 Third Century Professor in Computer Science Andrew Campbell
- Italian Briefs, Dartmouth Faculty Michael Wyatt

Skills

- Python, Stata, LaTeX, Slurm, Excel, Word